General Software Hints

Embedded Temp Sys

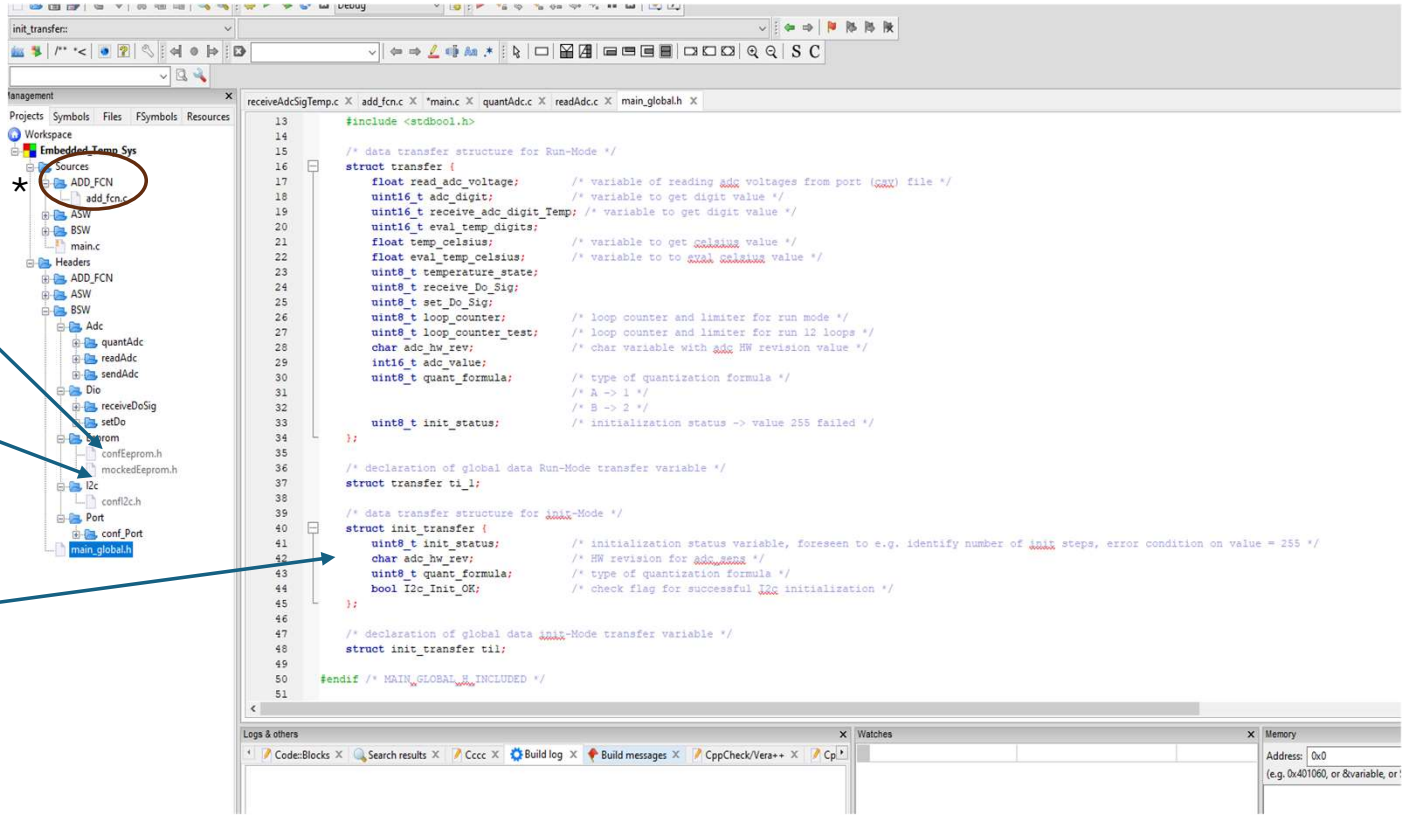
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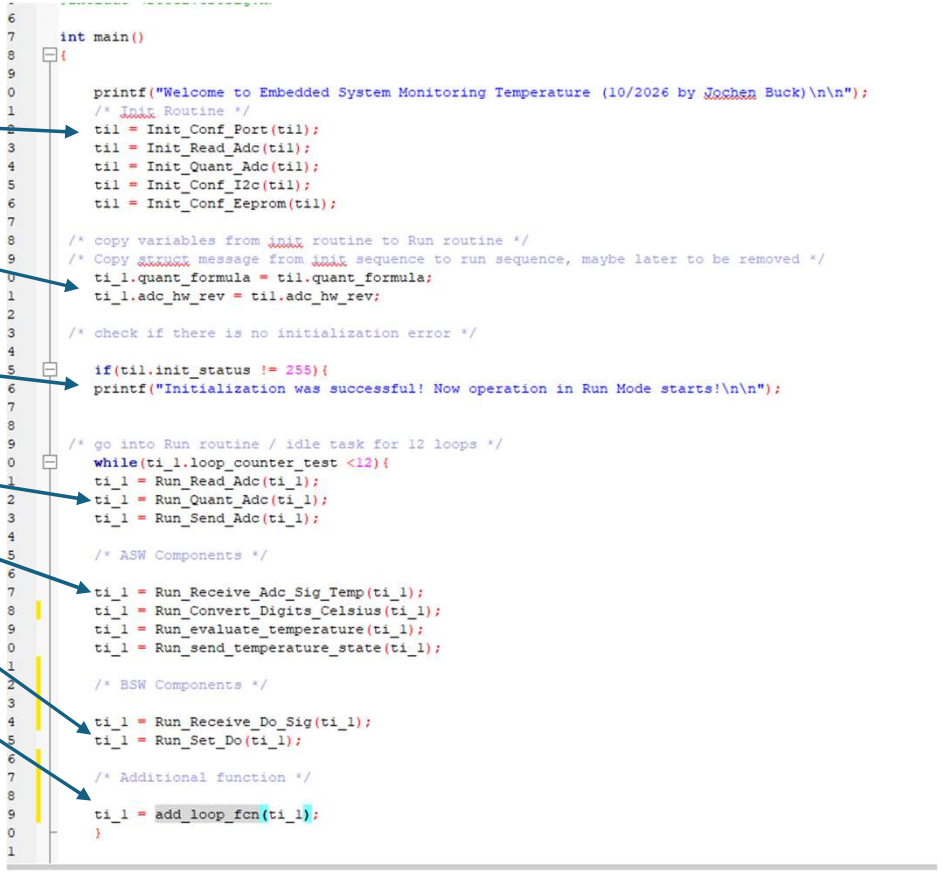
01 General Explanation sw architecture

- Basically, the SW – architecture is found similar in coding (BCs, FCs,ASW..)
- The functions are in the according BC Folders in Sub-folders in *.c and *.h files
- Data Communication is done via 2 global structs btw. SWCs
  -struct init_transfer (Init)
  -struct transfer(Run)
- In ADD_FCN is a special general counter which limits the execution to 12 Times *



02 Explanation Main Loop

- Main loop starts with Init section
- At the end of Init_Mode “hw_revision” and “quantization formula” selection is provided from init_struct to run_struct
- Before entering Run_Mode, init_status is checked for error
- In Run_Mode (while) is beginning with Input related functions (BSW), then Evaluation (ASW), at the End output Related functions (BSW)
- After output Related functions general loop counter is executed



```
int main()
{

    printf("Welcome to Embedded System Monitoring Temperature (10/2026 by Jochen Buck)\n\n");
    /* Init Routine */
    til = Init_Conf_Port(til);
    til = Init_Read_Adc(til);
    til = Init_Quant_Adc(til);
    til = Init_Conf_I2c(til);
    til = Init_Conf_Eeprom(til);

  /* copy variables from init routine to Run routine */
  /* Copy struct message from init sequence to run sequence, maybe later to be removed */
    ti_1.quant_formula = til.quant_formula;
    ti_1.adc_hw_rev = til.adc_hw_rev;

  /* check if there is no initialization error */

    if(til.init_status != 255){
    printf("Initialization was successful! Now operation in Run Mode starts!\n\n");


  /* go into Run routine / idle task for 12 loops */
    while(ti_1.loop_counter_test <12){
    ti_1 = Run_Read_Adc(ti_1);
    ti_1 = Run_Quant_Adc(ti_1);
    ti_1 = Run_Send_Adc(ti_1);

    /* ASW Components */

    ti_1 = Run_Receive_Adc_Sig_Temp(ti_1);
    ti_1 = Run_Convert_Digits_Celsius(ti_1);
    ti_1 = Run_evaluate_temperature(ti_1);
    ti_1 = Run_send_temperature_state(ti_1);

    /* BSW Components */

    ti_1 = Run_Receive_Do_Sig(ti_1);
    ti_1 = Run_Set_Do(ti_1);

    /* Additional function */

    ti_1 = add_loop_fcn(ti_1);
    }
```